

UNIT 5: MECHANISMS & PATTERNS OF EVOLUTION

5.3 – SPECIATION & MACROEVOLUTION

5.3 – Opening Question



Pinzón giant tortoise
(*C. duncanensis*)



Floreana giant tortoise
(*C. niger*)



Western Santa Cruz giant tortoise
(*C. porteri*)



Fernandina giant tortoise
(*C. phantasticus*)



Darwin Volcano giant tortoise
(*C. microphyes*)



Alcedo Volcano giant tortoise
(*C. vandenburghi*)



Sierra Negra giant tortoise
(*C. guntheri*)



Cerro Azul giant tortoise
(*C. vicina*)

- Examine the tortoises from Lesson 1 to the left.
- With a partner, determine what evidence could be used to make the case for each of the tortoises below to be grouped into their own unique species. How could a scientist verify each species as unique?

What exactly is a species?

- How New Species Evolve
- Modes of Speciation
- Principles of Macroevolution

How New Species Evolve

- **Macroevolution** is best observed within the fossil record.
 - ▣ Shows evolution on a large scale
- Involves **speciation**.
 - ▣ Splitting of one species into 2 or more new species
 - ▣ Macroevolution is the result of accumulation of microevolutions that result in formation of new species
- Every species has its own evolutionary history
 - ▣ **Species concept** – Refers to the different ways in which a species is defined
 - ▣ **Taxonomists** analyze and define species

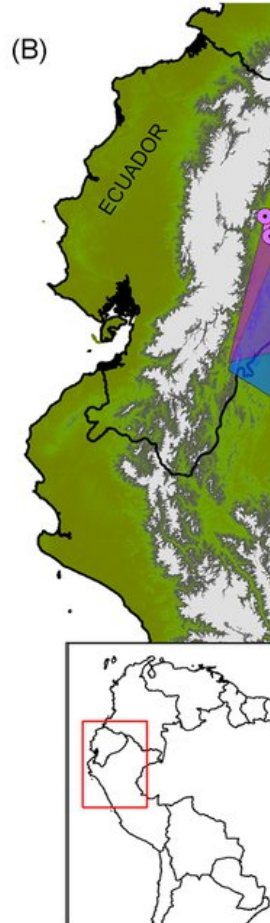
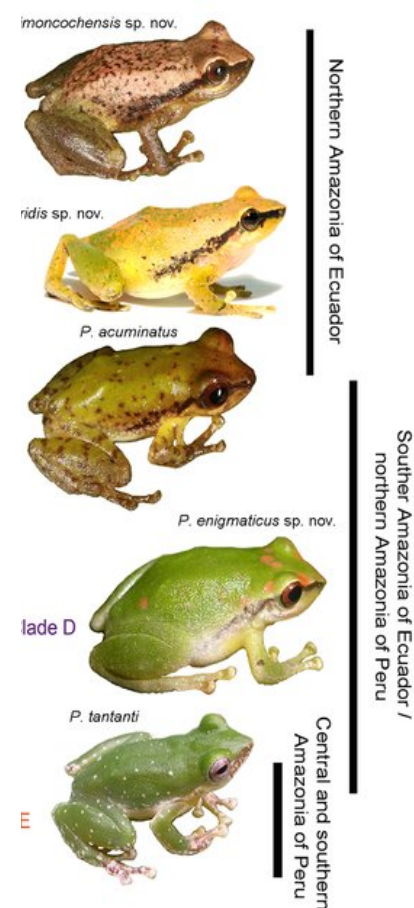
How New Species Evolve

□ Morphological Species Concept

- Based on analysis of **diagnostic traits** distinguishing one species from another.
- Can be distinguished anatomically
- Method used by Linnaeus
- Most species are described this way

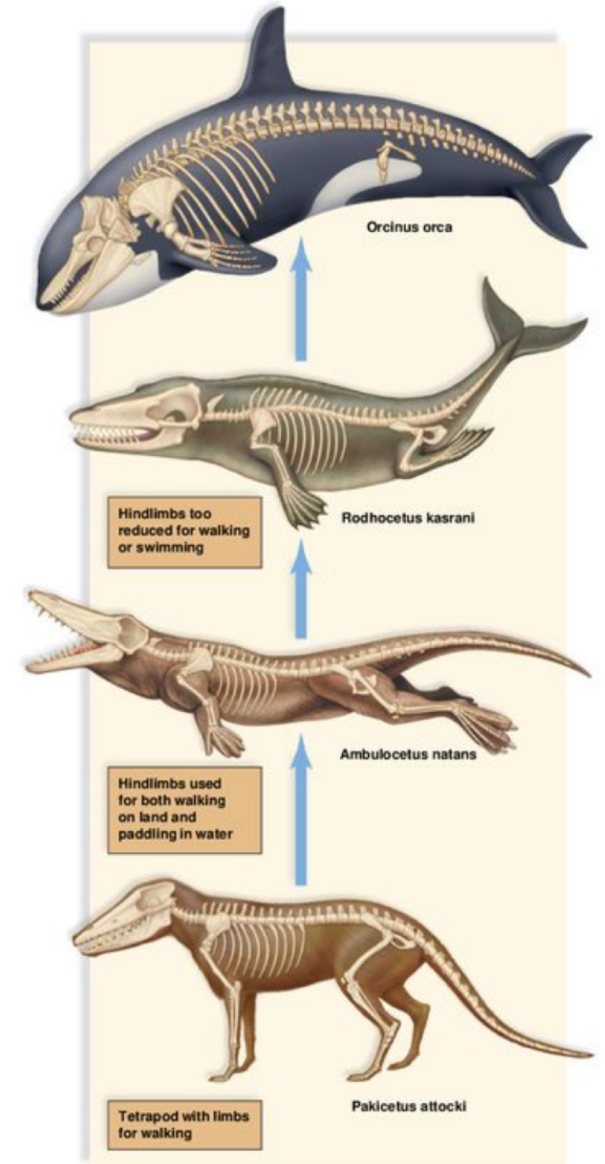
□ Drawbacks

- Some species, like different bacteria, have no observable, morphological traits to analyze
- **Cryptic species** – some species look so similar, it is difficult to identify them as distinct species.



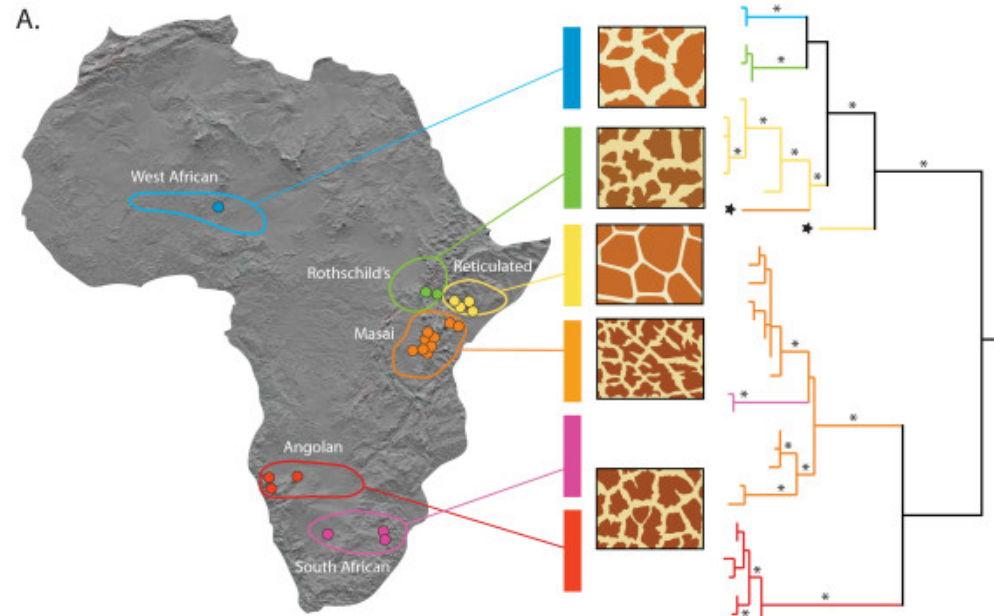
How New Species Evolve

- Evolutionary Species Concept
 - The evolutionary species concept distinguishes species from one another based on morphological (structural) traits
 - Critical traits for distinguishing species are called diagnostic traits
 - Members of a species must share the same, distinct, evolutionary pathway



How New Species Evolve

- Phylogenetic Species Concept
 - Relies on the identification of species based on common ancestry
 - Evolutionary “family tree”, or **phylogenetic tree** used
 - A branch that contains all descendants of a common ancestor is known as **monophyletic** – criterion used for defining a species
 - Advantage is it relies on DNA sequence, and not morphological traits



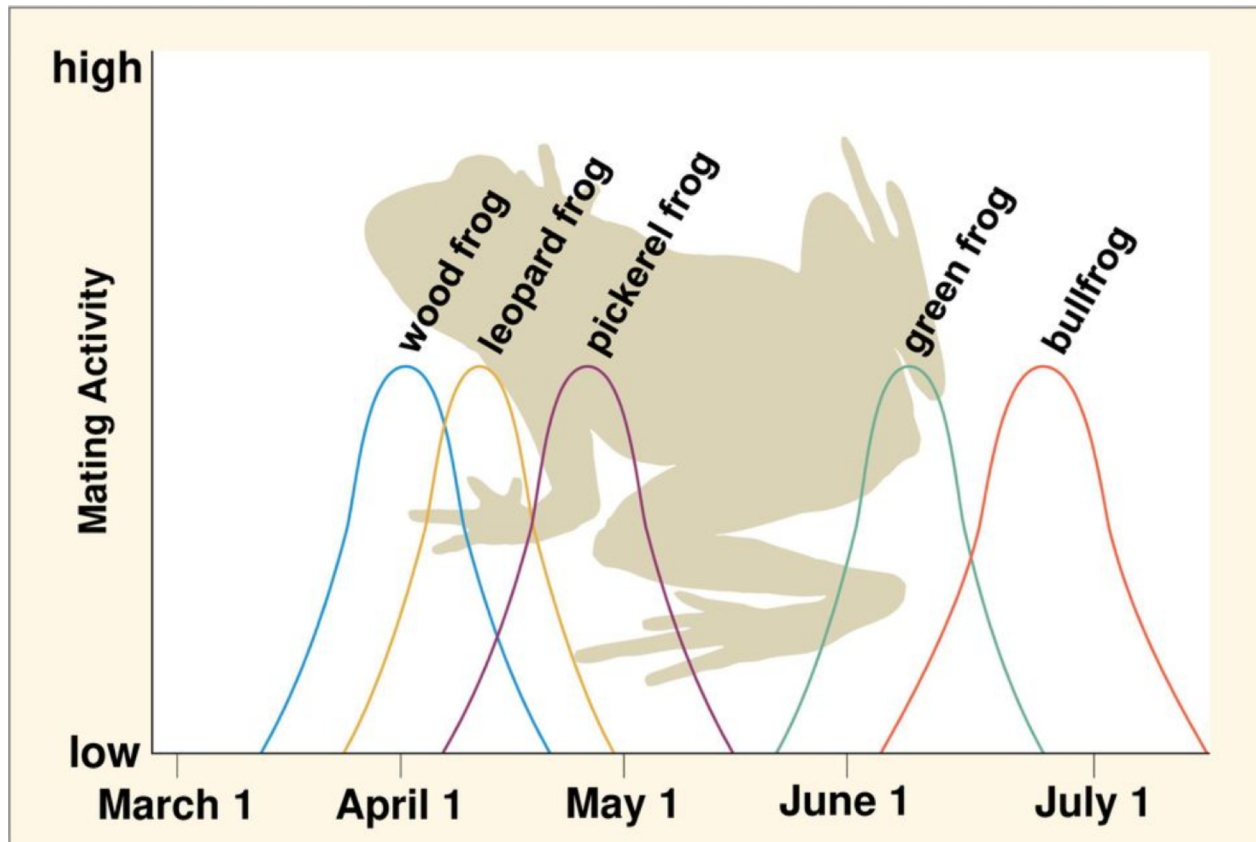
How New Species Evolve

- Biological Species Concept
 - ▣ Populations of the same species breed only among themselves
 - ▣ Experience **reproductive isolation** from other such populations
 - ▣ Often cannot be tested, because very few species in nature overlap in their distribution to even observe reproductive isolation

How New Species Evolve

- Reproductive isolating mechanisms inhibit gene flow between species - Two general types:
- **Prezygotic Isolating Mechanisms** – prevent mating attempts or make it unlikely that fertilization will be successful
 - **Habitat Isolation** - species occupy different habitats
 - **Temporal Isolation** - each reproduces at a different time
 - **Behavioral Isolation** - courtship patterns for recognizing mates
 - **Mechanical Isolation** - incompatible animal genitalia or plant floral structures
 - **Gamete Isolation** - gametes that meet do not fuse to become a zygote

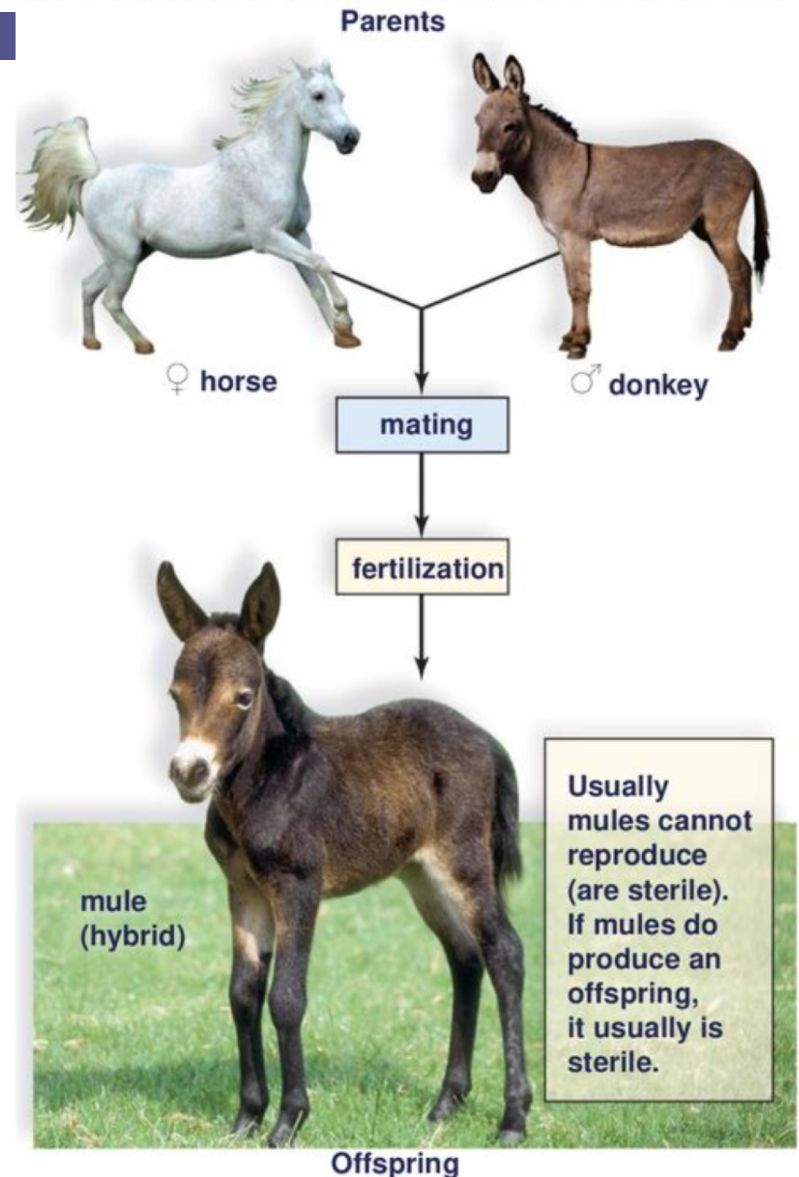
How New Species Evolve



Example of temporal isolation – Frogs from Ithica, New York above, have breeding period peaks that differ between species.

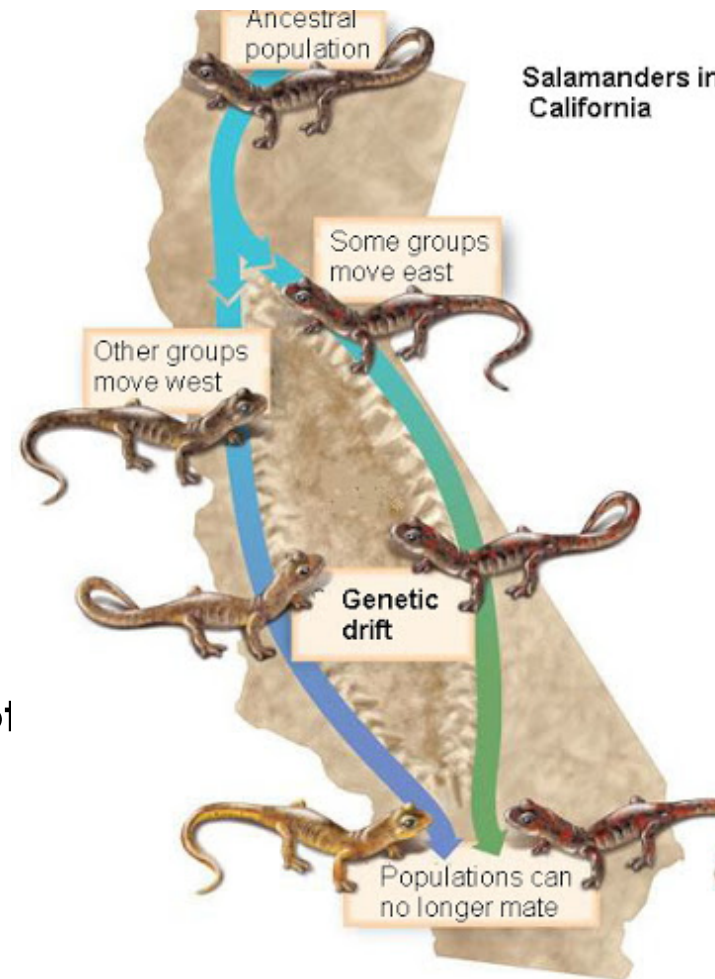
How New Species Evolve

- Reproductive Isolating Mechanisms (cont.)
- **Postzygotic Isolating Mechanisms** - Prevent hybrid offspring from developing or breeding
 - ▣ **Hybrid Inviability** - hybrid zygote is not viable and dies
 - ▣ **Hybrid Sterility** - hybrid zygote develops into a sterile adult



Modes of Speciation

- Researchers recognize 2 modes of speciation
 - One requires populations to be physically isolated from one another, and the other does not.
- Allopatric Speciation
 - Two geographically isolated populations of one species become different species over time
 - Can be due to differing selection pressures in differing environments
 - Reinforcement occurs when two populations come back together and no longer reproduce because they no longer recognize each other as mates due to changes



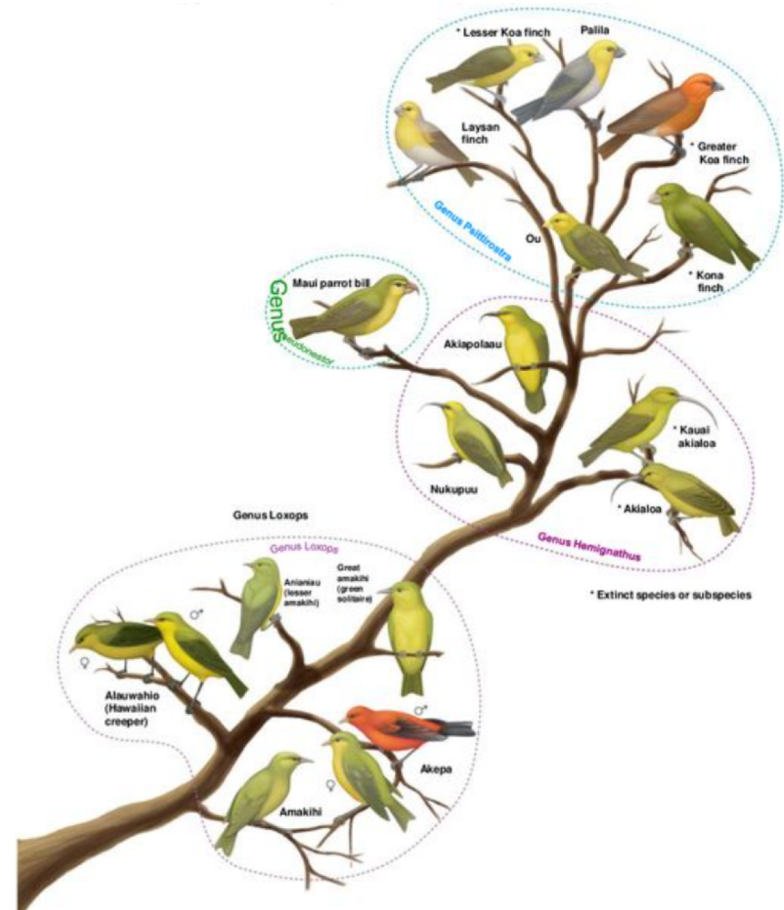
Modes of Speciation

□ Sympatric Speciation

- One population develops into two or more reproductively isolated groups - No prior geographic isolation
- In plants, sympatric speciation often involves **polyploidy** (more than the diploid $[2n]$ number)
- A polyploid plant can reproduce with itself, but cannot reproduce with the $2n$ population because not all the chromosomes would be able to pair during meiosis.
 - Two types of polyploidy are known:
 - Autopolyploidy - diploid plant produces diploid gametes due to nondisjunction during meiosis.
 - If diploid gamete fuses with a haploid gamete, a triploid plant results.
 - A triploid ($3n$) plant is sterile and cannot produce offspring
 - Allopolyploidy - more complicated process than autopolyploidy
 - Requires two different but related species of plants
 - Hybridization is followed by doubling of the chromosomes.

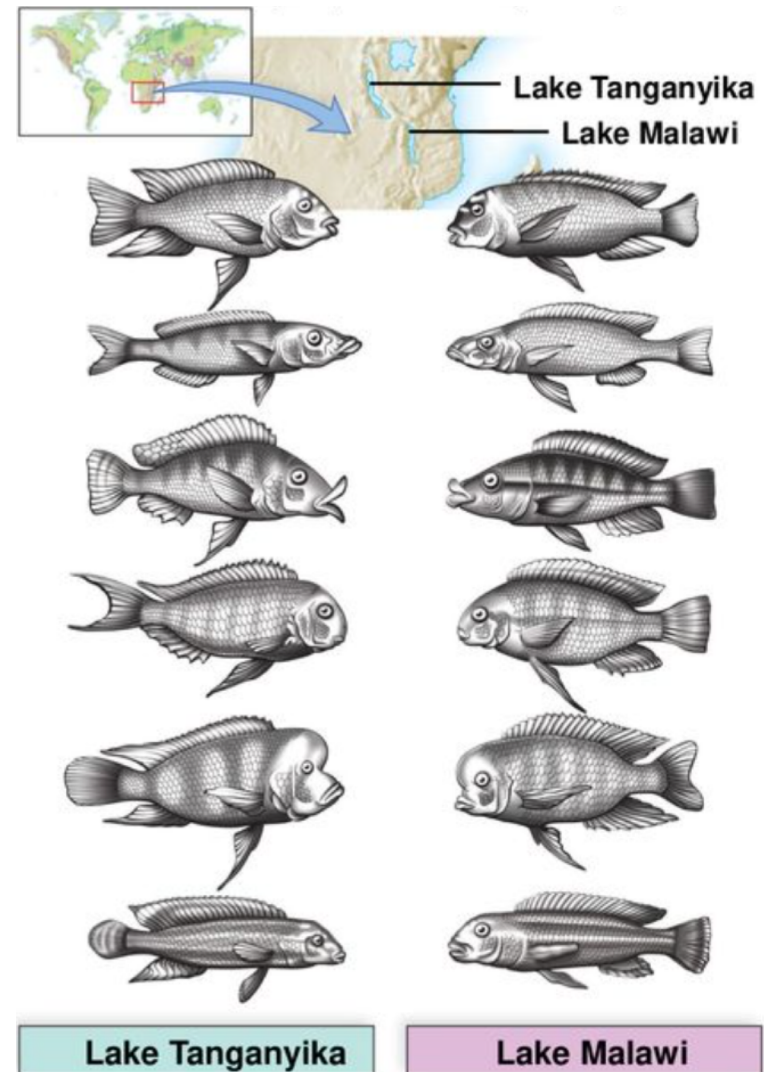
Modes of Speciation

- Adaptive Radiation
 - Occurs when members of a species invade several new geographically separate environments
 - The populations become adapted to the different environments
 - Many new species evolve from the single ancestral species
 - Darwin's finches are an example



Modes of Speciation

- Convergent Evolution
 - Occurs when a similar biological trait evolves in two unrelated species as a result of exposure to similar environments.
 - Traits evolving in this manner are termed analogous traits.
 - Similar function, but different origin
 - bird wing vs. bat wing
 - Chichlids in 2 African lakes
 - More closely related to species in own lake than morphological similar from other lake

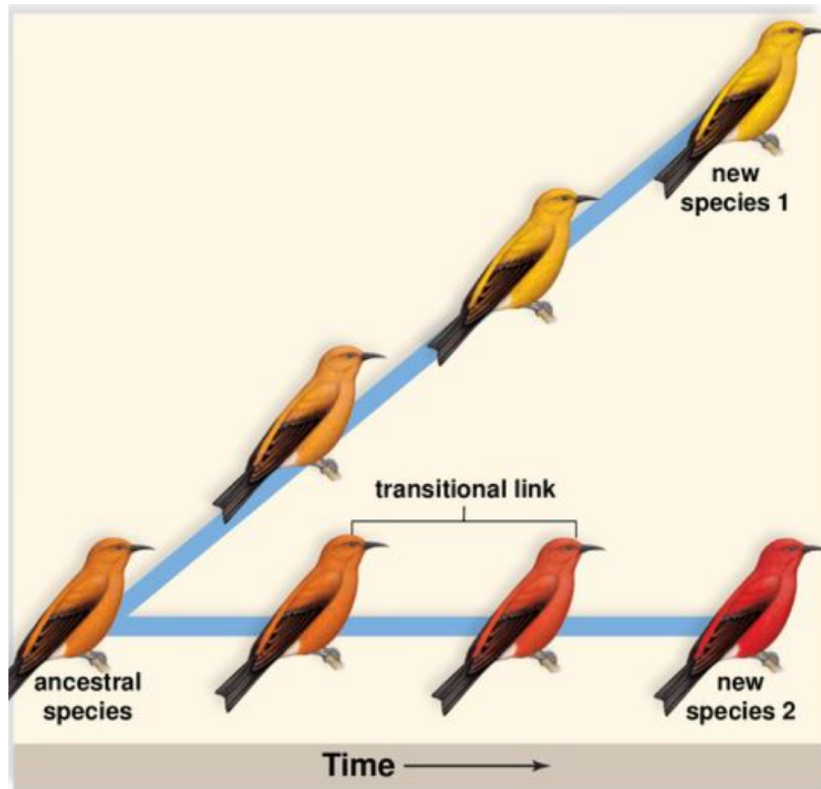


Principles of Macroevolution

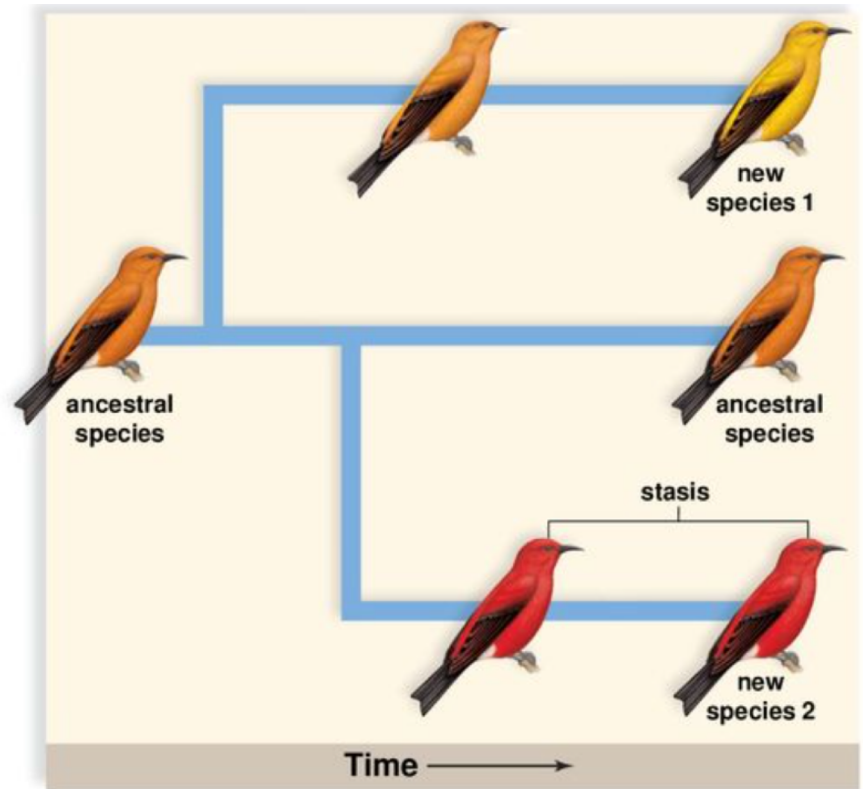
□ Macroevolution

- Evolution at the species or higher level of classification
- Some evolutionists support a **gradualistic model**
 - Evolution at the species level occurs gradually
 - Speciation occurs after populations become isolated
 - Each group continues its own evolutionary pathway
 - The gradualistic model suggests that it is difficult to indicate when speciation occurred
- Some paleontologists support the punctuated equilibrium model
 - Species can appear quite suddenly
 - Species then remain essentially unchanged phenotypically during a period of stasis (sameness) until they undergo extinction.
 - This model states that periods of equilibrium are punctuated by speciation

Principles of Macroevolution



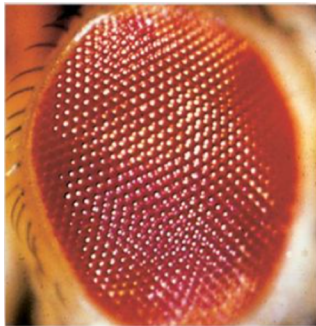
a. Gradualistic model



b. Punctuated equilibrium

Principles of Macroevolution

- Developmental Genes and Macroevolution
 - Genes can bring about radical changes in body shapes and organs.
 - Gene expression can influence development
 - A change in gene expression could stop a developmental process or continue it beyond its normal time.
 - Using modern technology, researchers discovered genes whose differential expression can bring about changes in body shapes and organs.
 - *Pax6* in eye development
 - *Tbx5* in limb development
 - *Hox* genes in development of overall shape



Principles of Macroevolution

- Macroevolution is not goal-oriented
 - ▣ The evolution of the horse (*Equus*)
 - Studied since the 1870s
 - Model for gradual, straight-line evolution with the modern-horse as its goal
 - Three trends were particularly evident during the evolution of the horse:
 - Increase in overall size
 - Toe reduction
 - Change in tooth size and shape
 - ▣ Discovery of more fossils has led to recognition that:
 - The lineage of a horse is complicated by the presence of many ancestors with varied traits
 - The direct ancestor of *Equus* is not known
 - Each ancestral species was adapted to its environment
 - Speciation, diversification, and extinction are common occurrences in the fossil record

Principles of Macroevolution

- Macroevolution is not goal-oriented
 - ▣ The evolution of the horse (Equus)

